



Zenith Popat

📞 Phone number: (+49) 15758085562 ✉ Email address: zenith.popat@gmail.com

🌐 Website: <https://zenithpopat.vercel.app/>

🌐 Website: <https://github.com/ZenithPopat>

🌐 Website: <https://www.linkedin.com/in/zenith-popat/>

📍 Home: Kaiserslautern (Germany)

WORK EXPERIENCE

🏢 **Rheinland-Pfälzische Technische Universität** – Kaiserslautern, Germany

Master Thesis Student

[01/02/2025 – Current]

- Designing and developing an **interactive web-based 3D scene editor** for large-scale **Gaussian Splatting datasets**, aimed at enabling visualization and manipulation of multiple point-based 3D representations within a single scene.
- Implemented **multi-file dataset loading and merging**, allowing independent Gaussian or PLY datasets to be combined into a unified scene while preserving per-object metadata and index boundaries.
- Developed **object-level interaction features**, including dataset selection, bounding box computation, metadata tracking, and foundations for transforms and visibility control.
- Optimized memory usage and rendering performance in **WebGL** by managing large Float32Array buffers, reducing redundant allocations, and minimizing scene update overhead.
- Bridged research concepts and production-grade engineering by translating academic Gaussian Splatting techniques into a usable, extensible web application built with **Babylon.js**.

🏢 **Everest Solutions GmbH** – Heidelberg, Germany

Full-Stack Developer (Working Student)

[04/2022 – 07/2024]

- Designed and implemented a **platform-wide GitHub-style tagging system**, enabling users to define custom tags and attach them across multiple ERP modules, significantly improving cross-module searchability and filtering.
- Owned the tagging system **end-to-end**, including data modeling, API design, frontend integration, and extensibility considerations, demonstrating architectural responsibility within a production SaaS platform.
- Designed and implemented **hierarchical relational data models** for departments, roles, and employees, accurately representing complex organizational structures in both database schema and UI.
- Developed and maintained **full-stack HR modules** using **TypeScript, React/Vue, Node.js, REST APIs, and PostgreSQL**, contributing directly to customer-facing ERP functionality.
- Improved overall platform stability and maintainability through **unit testing, systematic bug fixing, and collaborative Git-based development** within an agile engineering team.

🏢 **CibusCell Technology GmbH** – Speyer, Germany

Full-Stack Developer (Working Student)

[11/2021 – 03/2022]

- Contributed to an **early-stage SaaS cloud platform** for end-to-end management of hydrogen power plants, working primarily on frontend features with supporting backend integration.
- Onboarded by **identifying and resolving frontend and backend bugs** in a **React and Node.js** codebase, helping stabilize the platform during an early product phase.
- Designed and implemented **interactive dashboards** by integrating and visualizing operational data sourced from **Power BI**, improving accessibility of analytics for internal stakeholders.

- Collaborated closely within a small startup team, gaining exposure to **rapid iteration cycles, evolving requirements, and cross-functional product development**.
- Participated in early discussions around a planned **NestJS-based platform rewrite**, contributing to architectural planning and migration considerations and contributed in new project kickoff.

 **CodeTantra Tech Solutions Pvt. Ltd.** – Hyderabad, India

Product Development Engineer (Full-Time)

[12/2020 – 07/2021]

- Worked as a full-stack engineer on a **B2B EdTech SaaS platform** offering virtual classrooms, live coding sessions, online examinations, and remote proctoring for schools and universities.
- Onboarded by debugging and maintaining a **large production codebase** built with **Angular (JavaScript), Java, and Python**, improving reliability while gaining deep familiarity with the system architecture.
- Integrated **BigBlueButton WebRTC components** to enable real-time video classes, live meetings, and online examinations, supporting synchronous learning at scale.
- Designed and implemented **timed online examination modules**, handling exam **lifecycle management, session control, and submission constraints**.
- Built **anti-cheating mechanisms** to detect and prevent tab switching during exams, directly contributing to assessment integrity and institutional trust in the platform.

SKILLS

Typescript / React / JavaScript / Next.js / Angular.js / Express / Node.js / REST API / Redux / CSS / HTML / Tailwind / MySQL / SQL / MongoDB / NextAuth / Git / Zustand / Github / Prisma / Figma

PROJECTS

Personal Developer Portfolio

- Designed and built a **modern, responsive personal portfolio website** to showcase projects, experience, and technical skills in a clean and structured manner.
- Implemented a **component-driven React architecture** with Tailwind CSS for rapid styling, consistency, and responsiveness across devices.
- Integrated **lightweight 3D visuals and animations** using Three.js to enhance visual engagement without compromising performance or usability.

Link: <https://zenithpopat.vercel.app/>

Travel Blog (Full-Stack Web Application)

- Built a **full-stack travel blogging platform** using **Next.js and Prisma**, supporting authentication, content creation, dynamic routing, and relational data management.
- Implemented **secure authentication and media handling**, integrating third-party providers and cloud storage while addressing access control, upload limits, and production configuration issues.

EDUCATION AND TRAINING

Master in Computer Science

Rheinland-Pfälzische Technische Universität

City: Kaiserslautern | Country: Germany

Bachelor in Computer Science

Marwadi University

City: Rajkot | Country: India

ERASMUS STUDENT EXCHANGE

[30/08/2023 – 22/12/2023]

Erasmus Exchange Student (Master's Level) - University of Oulu - Oulu, Finland

- Designed and presented a **user-centered mobile application concept** for budget travelers, developing **paper prototypes and wireframes** and validating usability through structured HCI methods.
- Created a **unified budget-travel platform concept** combining accommodation, transport, and food discovery into a single application, reducing fragmentation across existing services.
- Co-authored a **survey-based research study** in Cyber Security exploring the use of **AR/VR/XR for network visualization** to improve penetration testing and early vulnerability detection.

[02/10/2022 – 01/03/2023]

Erasmus Exchange Student (Master's Level) - Technical University of Vienna (TU Wien) - Vienna, Austria

- Developed a **machine-learning-based Java agent** for the strategy game *Kalaha*, ranking **4th among 8–10 competing agents** in a multi-team evaluation.
- Conducted a **computer vision research project** on **eye-openness detection for in-cabin driver monitoring**, addressing driver distraction and drowsiness for traffic safety.
- Built **machine learning forecasting models** using the **LamaH climate dataset** to predict long-term climate attributes and assess environmental impact.

[24/06/2017 – 26/07/2017]

Erasmus Exchange Student (Summer School) - Rheinland-Pfälzische Technische Universität (RPTU) — Kaiserslautern, Germany

- Participated in an **Erasmus-funded academic exchange (summer school)** at RPTU, gaining early exposure to the **German and European higher education system**.
- Completed coursework in **Business Intelligence**, covering foundational concepts in data analysis and decision support, with practical **assignments implemented in Python**.

LANGUAGE SKILLS

Other language(s):

English

LISTENING C1 READING C1 WRITING C1

SPOKEN PRODUCTION C1 SPOKEN INTERACTION C1

German

LISTENING A1 READING A1 WRITING A1

SPOKEN PRODUCTION A1 SPOKEN INTERACTION A1

Spanish

LISTENING A1 READING A1 WRITING A1

SPOKEN PRODUCTION A1 SPOKEN INTERACTION A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user